

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Modular Unit 1

Adaptive Classified

OL-T39 · Standard and Compound Units

With Answers

Questions with full worked solutions

■ 27 pages

FAMILY MAP

Standard Compound Units

F01 Use pressure as a compound measure

1 Q | 3 marks

F02 Use speed, distance and time

10 Q | 30 marks

F03 Use density as a compound measure

1 Q | 2 marks

Question 05

4MA1-1H Q8

ORIGINAL QUESTION

3 marks

- 8 A cylinder is placed on a table.

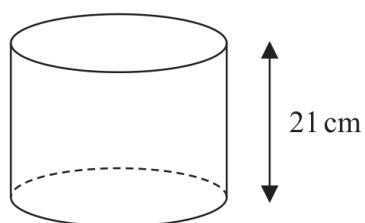


Diagram **NOT**
accurately drawn

The volume of the cylinder is 1575 cm^3

The force exerted by the cylinder on the table is 84 newtons.

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

Work out the pressure on the table due to the cylinder.

..... newtons/cm²

Worked Solution 05

Pressure

MODULAR UNIT 1

3 marks

Contact area

$$A = 1575 \div 21 = 75 \text{ cm}^2.$$

Pressure

$$84 \div 75 = 1.12.$$

Final answer

$$1.12 \text{ N/cm}^2$$

Question 17

4MA1-2H Q3

ORIGINAL QUESTION

3 marks

- 3 A plane takes 9 hours 36 minutes to fly from New Delhi to Perth.

The plane flies at an average speed of 820 km/h.

Work out the total distance the plane flies.

..... km

Worked Solution 17

Speed, distance and time

MODULAR UNIT 1

3 marks

Time

$$9 \text{ h } 36 \text{ min} = 9.6 \text{ h.}$$

Distance

$$820(9.6) = 7872.$$

Final answer

$$7872 \text{ km}$$

Question 20

4MA1-2H Q6

ORIGINAL QUESTION

3 marks

- 6 Change a speed of w metres per second to a speed in kilometres per hour.
Give your answer in terms of w in its simplest form.

..... kilometres per hour

Worked Solution 20

Convert speed units

MODULAR UNIT 1

3 marks

Convert

$$w \text{ m/s} = w \times 3600/1000 = 3.6w \text{ km/h.}$$

Final answer

$$3.6w \text{ km/h}$$

Question 16

4MA1-2H Q9

ORIGINAL QUESTION

3 marks

9 Change a speed of 27 kilometres per hour to a speed in metres per second.

..... m/s

Worked Solution 16

Convert kilometres per hour

MODULAR UNIT 1

3 marks

Convert distance and time

$$27 \text{ km} = 27\,000 \text{ m}, \quad 1 \text{ hour} = 3600 \text{ s}.$$

Divide

$$\frac{27\,000}{3600} = 7.5 \text{ m/s}.$$

Final answer

$$7.5 \text{ m/s}$$

Original question 16

4MA1-2H Q3 Q3

ORIGINAL QUESTION**3 marks**

- 3 An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

Worked solution 16

Chapter: Standard & Compound Units

MODULAR UNIT 1

3 marks

Q3 – Chapter: Standard & Compound Units

Family: Use speed, distance and time • Variant: Find average speed for one stage

Method $5 \text{ hours } 24 \text{ minutes} = 5 + 24/60 = 5.4 \text{ hours.}$ *Why: Convert minutes to hours before using kilometres per hour.***Method** $\text{Average speed} = 3980/5.4 = 737.037... \text{ kilometres per hour.}$ *Why: Average speed is distance divided by time.***Method**

To the nearest whole number, the speed is 737 kilometres per hour.

*Why: Round the unrounded quotient once at the end.***Final answer****speed:** 737 km/h

Original question 20

4MA1-2H Q7 Q7

ORIGINAL QUESTION**3 marks**

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

Worked solution 20

Chapter: Standard & Compound Units

MODULAR UNIT 1

3 marks

Q7 – Chapter: Standard & Compound Units

Family: Use speed, distance and time • Variant: Convert compound speed units

Method $81 \text{ km/h} = 81 \times 1000 \text{ metres per hour.}$ *Why: One kilometre is 1000 metres.***Method** $81 \times 1000 / (60 \times 60) = 22.5 \text{ metres per second.}$ *Why: One hour is 3600 seconds.***Final answer**

speed: 22.5 m/s

- 9 The density of gold is 19.3 g/cm^3
A gold bar has volume 150 cm^3

Work out the mass of the gold bar.

..... g

Worked solution

January 2020 | Question 9 | 2 marks

Family: Use density as a compound measure • Variant: Find mass from density and volume

Method / working

1. $\text{mass} = \text{density} \times \text{volume} = 19.3 \times 150.$

2. $\text{Mass} = 2895 \text{ g}.$

Final answer: 2895 g

10 Change a speed of 50 metres per second to a speed in kilometres per hour.

..... kilometres per hour

Worked solution

January 2020 | Question 10 | 3 marks

Family: Use speed, distance and time • Variant: Convert compound speed units

Method / working

1. Multiply m/s by 3.6 to convert to km/h.

2. $50 \times 3.6 = 180$.

3. Speed = 180 km/h.

Final answer: 180 km/h

- 1 A train takes 6 hours 39 minutes to travel from New Delhi to Kanpur.
The train travels a distance of 429 km.

Work out the average speed of the train.

Give your answer in km/h correct to one decimal place.

..... km/h

Worked solution

January 2021 | Question 1 | 3 marks

Family: Use speed, distance and time • Variant: Find average speed for one stage

Method / working

1. Convert 6 h 39 min to 6.65 hours.
 2. Average speed = $429 / 6.65 = 64.511\dots$ km/h.
 3. To 1 d.p., the speed is 64.5 km/h.
- Final answer: 64.5 km/h

- 1 A plane flew from Madrid to Dubai.

The distance the plane flew was 5658 km.

The flight time was 8 hours 12 minutes.

Work out the average speed of the plane.

..... km/h

Worked solution

June 2021 | Question 1 | 3 marks

Family: Use speed, distance and time • Variant: Find average speed for one stage

Method / working

1. Convert 8 h 12 min to 8.2 hours.
2. Use average speed = distance / time: $5658 / 8.2 = 690$.
3. State the speed with the requested km/h unit.

Final answer: 690 km/h

- 6 A rocket travelled 100 km at an average speed of 28 440 km/h.

Work out how long it took the rocket to travel the 100 km.

Give your answer in seconds, correct to the nearest second.

..... seconds

(Total for Question 6 is 3 marks)

Answer reference | May-Nov 2020 | Question 6

Family: Use speed, distance and time • Variant: Find average speed for one stage

6	$100 \div 28\,440 (= 0.0035\dots)$ or $28\,440 \div (60 \times 60) (= 7.9)$		3	M1
	'0.0035...' $\times 60 \times 60$ or $100 \div '7.9'$			M1
		13		A1 for 12.65 – 13
				Total 3 marks

Worked solution

May-Nov 2020 | Question 6 | 3 marks

Family: Use speed, distance and time | Variant: Find average speed for one stage

METHOD / WORKING

Worked solution

May-Nov 2020 | Question 6 | 3 marks

Family: Use speed, distance and time | Variant: Find average speed for one stage

Solution

Use

$$time = \frac{distance}{speed}$$

$$time = (100)/(28440) \text{ hours}$$

Convert hours to seconds:

$$(100)/(28440) \times 3600 = 12.658 \text{ dots}$$

To the nearest second,

$$12.658 \text{ dots approx } 13$$

Answer: 13 seconds.

FINAL ANSWER

- 2 Change a speed of 90 kilometres per hour to a speed in metres per second.
Show your working clearly.

..... m/s

(Total for Question 2 is 3 marks)

Answer reference | January 2023 | Question 2

Family: Use speed, distance and time • Variant: Find overall average speed across several stages

Question	Working	Answer	Mark	Notes	
2	$90 \times 1000 (=90\,000)$ or $\frac{90}{60 \times 60} (= 0.025 \text{ or } \frac{1}{40})$ or $\frac{1000}{60 \times 60} (= \frac{5}{18} = 0.277\dots)$ or sight of 1500		3	M1 For one of $\times 1000$ (eg sight of 90 000) or $(\div 60 \div 60)$ or $\div 3600$ oe ie correct conversion of distance units or of time units	M2 for $90 \div 3.6$ or $90 \times \frac{5}{18}$
	$\frac{90 \times 1000}{60 \times 60}$ oe eg $(1.5 \times 1000) \div 60$			M1 For a fully correct method with correct use of brackets eg $90\,000 \div 60 \times 60$ is M1 only if not recovered	
	<i>Working required</i>	25		A1 dep on M1	
Total 3 marks					

Worked solution

January 2023 | Question 2 | 3 marks

Family: Use speed, distance and time • Variant: Find overall average speed across several stages

Official final mark-scheme pages are embedded immediately after this question.